

CenterNet: Keypoint Triplets for Object Detection

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Abstract

In object detection, keypoint-based approaches often suffer a large number of incorrect object bounding boxes, arguably due to the lack of an additional look into the cropped regions. This paper presents an efficient solution which explores the visual patterns within each cropped region with minimal costs. We build our framework upon a representative one-stage keypoint-based detector named CornerNet. Our approach, named CenterNet, detects each object as a triplet, rather than a pair, of keypoints, which improves both precision and recall. Accordingly, we design two customized modules named cascade corner pooling and center pooling, which play the roles of enriching information collected by both top-left and bottom-right corners and providing more recognizable information at the central regions, respectively. On the MS-COCO dataset, CenterNet achieves an AP of **47.0%**, which outperforms all existing one-stage detectors by at least **4.9%**. Meanwhile, with a faster inference speed, CenterNet demonstrates quite comparable performance to the top-ranked two-stage detectors. Code is available at <https://github.com/Duankaiwen/CenterNet>.

1. Introduction

Object detection has been significantly improved and advanced with the help of deep learning, especially convolutional neural networks [11] (CNNs). In the current era, one of the most popular flowcharts is anchor-based [10, 12, 27, 31, 33], which placed a set of rectangles with pre-defined sizes, and regressed them to the desired place with the help of ground-truth objects. These approaches often need a large number of anchors to ensure a sufficiently high IoU (intersection over union) rate with the ground-truth objects, and the size and aspect ratio of each anchor box need to be manually designed. In addition, anchors are usually not

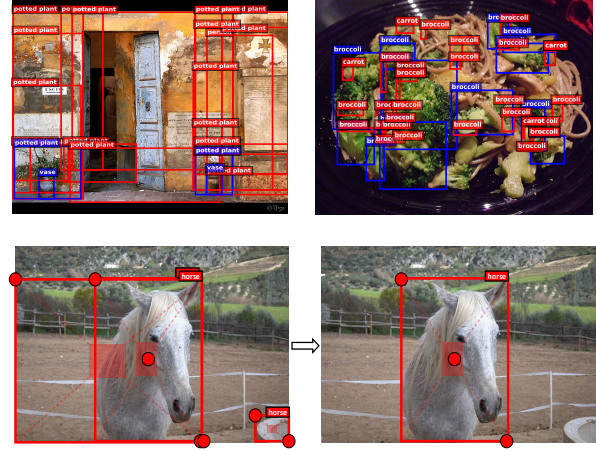


Figure 1: In the first row, we visualize the top 100 bounding boxes (according to the MS-COCO dataset standard) of CornerNet. Ground-truth and predicted objects are marked in blue and red, respectively. In the second row, we show that correct predictions can be determined by checking the central parts.

aligned with the ground-truth boxes, which is not conducive to the bounding box classification task.

To overcome the drawbacks of anchor-based approaches, a keypoint-based object detection pipeline named CornerNet [20] was proposed. It represented each object by a pair of corner keypoints, which bypassed the need of anchor boxes and achieved the state-of-the-art one-stage object detection accuracy. Nevertheless, the performance of CornerNet is still restricted by its relatively weak ability of referring to the global information of an object. That is to say, since each object is constructed by a pair of corners, the algorithm is sensitive to detect the boundary of objects, meanwhile not being aware of which pairs of keypoints should be grouped into objects. Consequently, as shown in Figure 1, it often generates some incorrect bounding boxes, most of which could be easily filtered out with complementary information, e.g., the aspect ratio.

*This work was done when the first author was interning at Huawei Noah's Ark Lab.

CenterNet：基于关键点三元组的目标检测

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摘要

In object detection, keypoint-based approaches often suffer a large number of incorrect object bounding boxes, arguably due to the lack of an additional look into the cropped regions. This paper presents an efficient solution which explores the visual patterns within each cropped region with minimal costs. We build our framework upon a representative one-stage keypoint-based detector named CornerNet. Our approach, named CenterNet, detects each object as a triplet, rather than a pair, of keypoints, which improves both precision and recall. Accordingly, we design two customized modules named cascade corner pooling and center pooling, which play the roles of enriching information collected by both top-left and bottom-right corners and providing more recognizable information at the central regions, respectively. On the MS-COCO dataset, CenterNet achieves an AP of 47.0%, which outperforms all existing one-stage detectors by at least 4.9%. Meanwhile, with a faster inference speed, CenterNet demonstrates quite comparable performance to the top-ranked two-stage detectors. Code is available at <https://github.com/Duankaiwen/CenterNet>.

1. 引言

目标检测在深度学习，尤其是卷积神经网络[11]（CNNs）的推动下取得了显著改进与进步。当前阶段，最主流的流程框架之一是基于锚点的[10, 12, 27, 31, 33]方法，这类方法通过预设一组尺寸固定的矩形框，并借助真实标注物体将这些锚框回归到目标位置。此类方法通常需要大量锚点以确保与真实标注物体保持足够高的交并比（IoU），且每个锚框的尺寸和宽高比需人工设计。此外，锚框通常无法

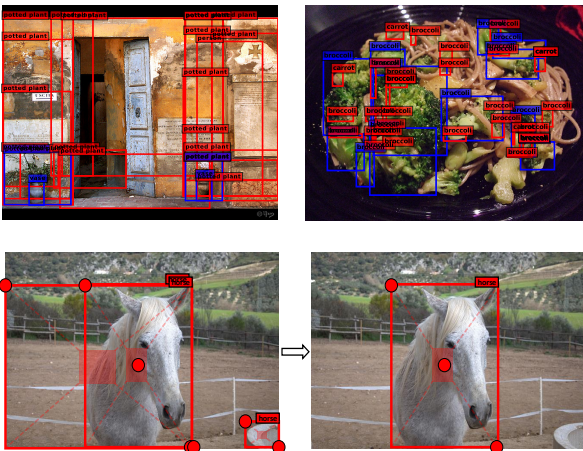


图1：在第一行，我们可视化了CornerNet的前100个边界框（依据MS-COCO数据集标准）。真实标注与预测目标分别以蓝色和红色标出。在第二行，我们展示了通过检查中心区域可以确定正确预测结果。

与真实边界框对齐，这不利于边界框分类任务。

为了克服基于锚点方法的缺点，一种名为Corner-Net [20]的基于关键点目标检测流程被提出。它通过一对角点关键点来表示每个目标，从而绕过了对锚框的需求，并实现了一阶段目标检测的最先进精度。然而，Corner-Net的性能仍受限于其相对较弱的参考目标全局信息的能力。也就是说，由于每个目标由一对角点构建，该算法对目标边界的检测较为敏感，同时无法感知哪些关键点对应组合成目标。因此，如图1所示，它常常生成一些错误的边界框，其中大部分可以通过补充信息*e.g.*（即宽高比）轻松过滤掉。

*This work was done when the first author was interning at Huawei Noah's Ark Lab.

To address this issue, we equip CornerNet with an ability of perceiving the visual patterns within each proposed region, so that it can identify the correctness of each bounding box by itself. In this paper, we present a low-cost yet effective solution named **CenterNet**, which explores the central part of a proposal, *i.e.*, the region that is close to the geometric center, with one extra keypoint. Our intuition is that, if a predicted bounding box has a high IoU with the ground-truth box, then the probability that the center keypoint in its central region is predicted as the same class is high, and vice versa. Thus, during inference, after a proposal is generated as a pair of corner keypoints, we determine if the proposal is indeed an object by checking if there is a center keypoint of the same class falling within its central region. The idea, as shown in Figure 1, is to use a triplet, instead of a pair, of keypoints to represent each object.

Accordingly, for better detecting center keypoints and corners, we propose two strategies to enrich center and corner information, respectively. The first strategy is named **center pooling**, which is used in the branch for predicting center keypoints. Center pooling helps the center keypoints obtain more recognizable visual patterns within objects, which makes it easier to perceive the central part of a proposal. We achieve this by getting out the max summed response in both horizontal and vertical directions of the center keypoint on a feature map for predicting center keypoints. The second strategy is named **cascade corner pooling**, which equips the original corner pooling module [20] with the ability of perceiving internal information. We achieve this by getting out the max summed response in both boundary and internal directions of objects on a feature map for predicting corners. Empirically, we verify that such a two-directional pooling method is more stable, *i.e.*, being more robust to feature-level noises, which contributes to the improvement of both precision and recall.

We evaluate the proposed CenterNet on the MS-COCO dataset [25], one of the most popular benchmarks for large-scale object detection. CenterNet, with both center pooling and cascade corner pooling incorporated, reports an AP of 47.0% on the test-dev set, which outperforms all existing one-stage detectors by a large margin. With an average inference time of 270ms using a 52-layer hourglass backbone [29] and 340ms using a 104-layer hourglass backbone [29] per image, CenterNet is quite efficient yet closely matches the state-of-the-art performance of the other two-stage detectors.

The remainder of this paper is organized as follows. Section 2 briefly reviews related work, and Section 3 details the proposed CenterNet. Experimental results are given in Section 4, followed by the conclusion in Section 5.

2. Related Work

Object detection involves locating and classifying the objects. In the deep learning era, powered by deep convolutional neural networks, object detection approaches can be roughly categorized into two main types of pipelines, namely, two-stage approaches and one-stage approaches.

Two-stage approaches divide the object detection task into two stages: extract RoIs, then classify and regress the RoIs.

R-CNN [11] uses a selective search method [42] to locate RoIs in the input images and uses a DCN-based regionwise classifier to classify the RoIs independently. SPP-Net [13] and Fast-RCNN [10] improve R-CNN by extracting the RoIs from the feature maps. Faster-RCNN [33] is allowed to be trained end to end by introducing RPN (region proposal network). RPN can generate RoIs by regressing the anchor boxes. Later, the anchor boxes are widely used in the object detection task. Mask-RCNN [12] adds a mask prediction branch on the Faster-RCNN, which can detect objects and predict their masks at the same time. R-FCN [6] replaces the fully connected layers with the position-sensitive score maps for better detecting objects. Cascade R-CNN [4] addresses the problem of overfitting at training and quality mismatch at inference by training a sequence of detectors with increasing IoU thresholds. The keypoint-based object detection approaches [40, 28] are proposed to avoid the disadvantages of using anchor boxes and bounding boxes regression. Other meaningful works are proposed for different problems in object detection, *e.g.*, [47, 21] focus on the architecture design, [1, 9, 36, 44] focus on the contextual relationship, [22, 3] focus on the multi-scale unification.

One-stage approaches remove the RoI extraction process and directly classify and regress the candidate anchor boxes.

YOLO [31] uses fewer anchor boxes (divide the input image into an $S \times S$ grid) to do regression and classification. YOLOv2 [32] improves the performance by using more anchor boxes and a new bounding box regression method. SSD [27] places anchor boxes densely over an input image and use features from different convolutional layers to regress and classify the anchor boxes. DSSD [8] introduces a deconvolution module into SSD to combine low-level and high-level features. While R-SSD [17] uses pooling and deconvolution operations in different feature layers to combine low-level and high-level features. RON [19] proposes a reverse connection and an objectness prior to extract multiscale features effectively. RefineDet [45] refines the locations and sizes of the anchor boxes for two times, which inherits the merits of both one-stage and two-stage approaches. CornerNet [20] is another keypoint-based approach, which directly detects an object using a pair of corners. Although CornerNet achieves high performance, it still has more room to improve.

为了解决这个问题，我们为CornerNet配备了感知每个候选区域内视觉模式的能力，使其能够自行判断每个边界框的正确性。本文提出了一种低成本且高效的解决方案——CenterNet，它通过一个额外的关键点来探索候选区域的中心部分，即靠近几何中心的区域*i.e.*。我们的直觉是，如果一个预测的边界框与真实框具有较高的IoU，那么其中心区域内的中心关键点被预测为相同类别的概率也很高，反之亦然。因此，在推理过程中，当候选框由一对角点关键点生成后，我们通过检查其中心区域内是否存在相同类别的中心关键点，来判断该候选框是否确实是一个物体。如图1所示，这一思路是使用三个关键点（而非两个）来表示每个物体。

因此，为了更好地检测中心关键点和角点，我们提出了两种策略来分别丰富中心和角点信息。第一种策略称为中心池化，应用于预测中心关键点的分支中。中心池化帮助中心关键点在物体内部获取更具辨识度的视觉模式，从而更容易感知候选框的中心部分。我们通过在预测中心关键点的特征图上，沿水平和垂直方向提取最大总和响应来实现这一点。第二种策略称为级联角点池化，它使原始角点池化模块[20]具备了感知内部信息的能力。我们通过在预测角点的特征图上，沿物体边界和内部方向提取最大总和响应来实现这一点。经验表明，这种双向池化方法更加稳定，*i.e.*，对特征级噪声具有更强的鲁棒性，这有助于提高精确率和召回率。

我们在MS-COCO数据集[25]上评估了所提出的CenterNet，该数据集是大规模目标检测最流行的基准之一。结合了中心池化和级联角点池化的CenterNet在测试开发集上报告了**47.0%**的AP，大幅超越了所有现有的一阶段检测器。使用52层沙漏骨干网络[29]时，平均推理时间为每张图像270毫秒；使用104层沙漏骨干网络[29]时，为340毫秒。CenterNet效率极高，同时与其他两阶段检测器的最先进性能表现相当接近。

本文的其余部分组织如下。第2节简要回顾了相关工作，第3节详细介绍了提出的CenterNet。实验结果在第4节给出，第5节为结论。

2. 相关工作

目标检测涉及定位和分类物体。在深度学习时代，得益于深度卷积神经网络的推动，目标检测方法大致可分为两种主要流程类型，即两阶段方法和单阶段方法。

两阶段方法将目标检测任务分为两个阶段：提取感兴趣区域（RoIs），然后对这些RoIs进行分类和回归。

R-CNN [11]采用选择性搜索方法[42]在输入图像中定位感兴趣区域（RoIs），并利用基于深度卷积网络的区域分类器对RoIs进行独立分类。SPP-Net [13]和Fast-RCNN [10]通过从特征图中提取RoIs改进了R-CNN。Faster-RCNN [33]通过引入区域提议网络（RPN）实现了端到端训练，RPN可通过回归锚框生成RoIs。此后，锚框在目标检测任务中得到广泛应用。Mask-RCNN [12]在Faster-RCNN基础上增加了掩码预测分支，能同时检测目标并预测其掩码。R-FCN [6]用位置敏感得分图替代全连接层，以提升目标检测性能。Cascade R-CNN [4]通过使用递增IoU阈值训练级联检测器，解决了训练过拟合与推理质量不匹配的问题。基于关键点的目标检测方法[40, 28]被提出以规避锚框与边界框回归的缺陷。其他针对目标检测中不同问题的研究也具有重要意义：*e.g.*，[47, 21]专注于架构设计，[1, 9, 36, 44]聚焦上下文关系，[22, 3]着力于多尺度统一。

单阶段方法移除了RoI提取过程，直接对候选锚框进行分类和回归。

YOLO [31]采用较少的锚框（将输入图像划分为 $S \times S$ 网格）进行回归与分类。YOLOv2 [32]通过使用更多锚框及新的边界框回归方法提升了性能。SSD [27]在输入图像上密集布置锚框，并利用不同卷积层的特征对锚框进行回归和分类。DSSD [8]在SSD中引入反卷积模块以融合低层与高层特征。而R-SSD [17]则在不同特征层中使用池化与反卷积操作来结合低层与高层特征。RON [19]提出反向连接机制和物体性先验，以有效提取多尺度特征。RefineDet [45]对锚框的位置和尺寸进行两次细化，兼具单阶段与两阶段方法的优势。CornerNet [20]是另一种基于关键点的方法，直接通过一对角点检测物体。尽管CornerNet取得了优异性能，其仍有较大改进空间。

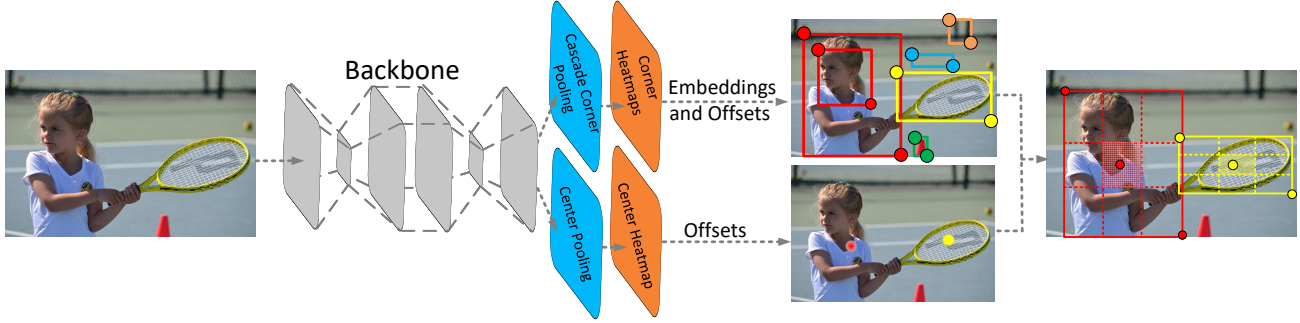


Figure 2: Architecture of CenterNet. A convolutional backbone network applies cascade corner pooling and center pooling to output two corner heatmaps and a center keypoint heatmap, respectively. Similar to CornerNet, a pair of detected corners and the similar embeddings are used to detect a potential bounding box. Then the detected center keypoints are used to determine the final bounding boxes.

Method	FD	FD ₅	FD ₂₅	FD ₅₀	FD _S	FD _M	FD _L
CornerNet	37.8	32.7	36.8	43.8	60.3	33.2	25.1

Table 1: False discovery rates (%) of CornerNet. The false discovery rate reflects the distribution of incorrect bounding boxes. The results suggest the incorrect bounding boxes account for a large proportion.

3. Our Approach

3.1. Baseline and Motivation

This paper uses CornerNet [20] as the baseline. For detecting corners, CornerNet produces two heatmaps: a heatmap of top-left corners and a heatmap of bottom-right corners. The heatmaps represent the locations of keypoints of different categories and assigns a confidence score for each keypoint. Besides, it also predicts an embedding and a group of offsets for each corner. The embeddings are used to identify if two corners are from the same object. The offsets learn to remap the corners from the heatmaps to the input image. For generating object bounding boxes, top- k left-top corners and bottom-right corners are selected from the heatmaps according to their scores, respectively. Then, the distance of the embedding vectors of a pair of corners is calculated to determine if the paired corners belong to the same object. An object bounding box is generated if the distance is less than a threshold. The bounding box is assigned a confidence score, which equals to the average scores of the corner pair.

In Table 1, we provide a deeper analysis of CornerNet. We count the FD¹ (false discovery) rate of CornerNet on the MS-COCO validation dataset, defined as the proportion of the incorrect bounding boxes. The quantitative re-

sults demonstrate the incorrect bounding boxes account for a large proportion even at low IoU thresholds, *e.g.*, CornerNet obtains 32.7% FD rate at IoU = 0.05. This means in average, 32.7 out of every 100 object bounding boxes have IoU lower than 0.05 with the ground-truth. The small incorrect bounding boxes are even more, which achieves 60.3% FD rate. One of the possible reasons lies in that CornerNet cannot look into the regions inside the bounding boxes. To make CornerNet [20] perceive the visual patterns in bounding boxes, one potential solution is to adapt CornerNet into a two-stage detector, which uses the RoI pooling [10] to look into the visual patterns in bounding boxes. However, it is known that such a paradigm is computationally expensive.

In this paper, we propose a highly efficient alternative called **CenterNet** to explore the visual patterns within each bounding box. For detecting an object, our approach uses a triplet, rather than a pair, of keypoints. By doing so, our approach is still a one-stage detector, but partially inherits the functionality of RoI pooling. Our approach only pays attention to the center information, the cost of our approach is minimal. Meanwhile, we further introduce the visual patterns within objects into the keypoint detection process by using center pooling and cascade corner pooling.

3.2. Object Detection as Keypoint Triplets

The overall network architecture is shown in Figure 2. We represent each object by a center keypoint and a pair of corners. Specifically, we embed a heatmap for the center keypoints on the basis of CornerNet and predict the offsets of the center keypoints. Then, we use the method proposed in CornerNet [20] to generate top- k bounding boxes. However, to effectively filter out the incorrect bounding boxes, we leverage the detected center keypoints and resort to the following procedure: (1) select top- k center keypoints according to their scores; (2) use the corresponding offsets to remap these center keypoints to the input image; (3) de-

¹FD = 1 - AP, where AP denotes the average precision at IoU = [0.05 : 0.05 : 0.5] on the MS-COCO dataset. Also, FD _{i} = 1 - AP _{i} , where AP _{i} denotes the average precision at IoU = $i/100$, FD_{scale} = 1 - AP_{scale}, where scale = {small, medium, large}, denotes the scale of object.

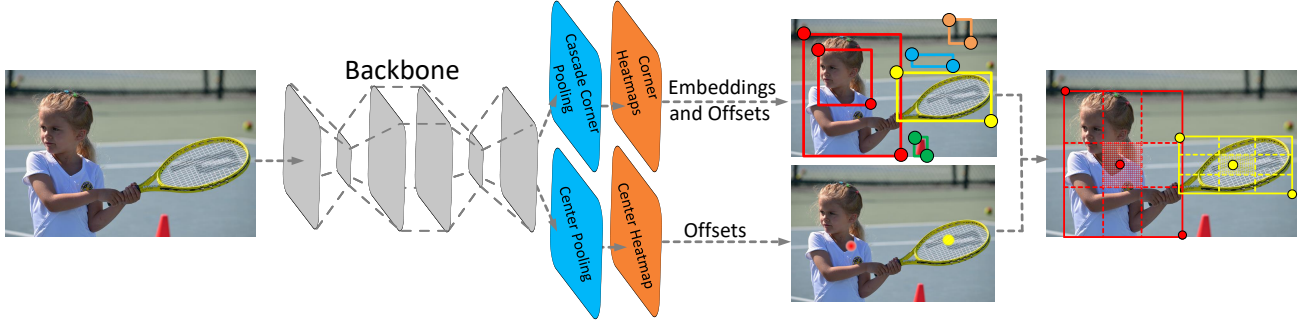


图2: CenterNet架构。卷积骨干网络应用级联角点池化和中心池化, 分别输出两个角点热图和一个中心关键点热图。与CornerNet类似, 通过检测到的一对角点及其相似嵌入向量来检测潜在边界框。随后, 利用检测到的中心关键点来确定最终边界框。

Method	FD	FD ₅	FD ₂₅	FD ₅₀	FD _S	FD _M	FD _L
CornerNet	37.8	32.7	36.8	43.8	60.3	33.2	25.1

表1: CornerNet的误检率(%)。误检率反映了错误边界框的分布情况。结果表明, 错误边界框占比较大。

3. 我们的方法

3.1. 基线方法与动机

本文以CornerNet [20]为基线模型。在角点检测方面, CornerNet会生成两个热力图: 左上角点热力图与右下角点热力图。这些热力图表征了不同类别关键点的位置, 并为每个关键点分配置信度分数。此外, 模型还会预测每个角点的嵌入向量及一组偏移量。嵌入向量用于判断两个角点是否属于同一物体, 偏移量则用于将热力图中的角点位置重映射至输入图像。在生成物体边界框时, 系统会依据得分分别从热力图中选取前 $\{v^*\}$ 个左上角点与右下角点。随后计算成对角点的嵌入向量距离, 以判定配对角点是否属于同一物体。若该距离小于设定阈值, 则生成物体边界框, 其置信度分数取为该对角点得分的平均值。

在表1中, 我们对CornerNet进行了更深入的分析。我们统计了CornerNet在MS-COCO验证数据集上的FD¹ (误报)率, 其定义为错误边界框所占比例。定量结

¹FD = 1 - AP, where AP denotes the average precision at IoU = [0.05 : 0.05 : 0.5] on the MS-COCO dataset. Also, FD_{*i*} = 1 - AP_{*i*}, where AP_{*i*} denotes the average precision at IoU = $i/100$, FD_{scale} = 1 - AP_{scale}, where scale = {small, medium, large}, denotes the scale of object.

结果表明, 即使在较低的IoU阈值下, 错误的边界框也占据了很大比例, *e.g.* CornerNet在IoU为=0.05时获得了32.7%的FD率。这意味着平均每100个物体边界框中, 有32.7个与真实标注的IoU低于0.05。其中较小的错误边界框更多, FD率达到了60.3%。可能的原因之一是CornerNet无法观察边界框内部区域。为了让CornerNet[20]感知边界框内的视觉模式, 一种潜在的解决方案是将CornerNet改进为两阶段检测器, 利用RoI池化[10]来观察边界框内的视觉模式。然而, 众所周知, 这种范式计算成本高昂。

本文提出了一种名为CenterNet的高效替代方案, 旨在探索每个边界框内的视觉模式。对于物体检测, 我们的方法使用三个关键点而非一对关键点。通过这种方式, 我们的方法仍是一个单阶段检测器, 但部分继承了RoI池化的功能。我们的方法仅关注中心信息, 因此成本极低。同时, 我们通过引入中心池化和级联角点池化, 进一步将物体内部的视觉模式融入关键点检测过程中。

3.2. 将目标检测视为关键点三元组

整体网络架构如图2所示。我们通过一个中心关键点和一对角点来表示每个物体。具体而言, 我们在CornerNet的基础上嵌入中心关键点的热力图, 并预测中心关键点的偏移量。随后, 我们采用CornerNet[20]提出的方法生成前 k 个边界框。然而, 为有效滤除不正确的边界框, 我们利用检测到的中心关键点并采用以下流程: (1) 根据得分选取前 k 个中心关键点; (2) 使用对应偏移量将这些中心关键点重映射至输入图像; (3) 删

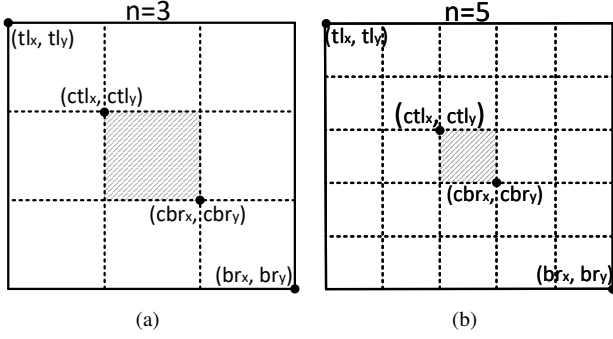


Figure 3: (a) The central region when $n = 3$. (b) The central region when $n = 5$. The solid rectangles denote the predicted bounding boxes and the shaded regions denote the central regions.

fine a central region for each bounding box and check if the central region contains center keypoints. Note that the class labels of the checked center keypoints should be same as that of the bounding box; (4) if a center keypoint is detected in the central region, we will preserve the bounding box. The score of the bounding box will be replaced by the average scores of the three points, *i.e.*, the top-left corner, the bottom-right corner and the center keypoint. If there are no center keypoints detected in its central region, the bounding box will be removed.

The size of the central region in the bounding box affects the detection results. For example, smaller central regions lead to a low recall rate for small bounding boxes, while larger central regions lead to a low precision for large bounding boxes. Therefore, we propose a scale-aware central region to adaptively fit the size of bounding boxes. The scale-aware central region tends to generate a relatively large central region for a small bounding box, while a relatively small central region for a large bounding box. Suppose we want to determine if a bounding box i needs to be preserved. Let tl_x and tl_y denote the coordinates of the top-left corner of i and br_x and br_y denote the coordinates of the bottom-right corner of i . Define a central region j . Let ctl_x and ctl_y denote the coordinates of the top-left corner of j and cbr_x and cbr_y denote the coordinates of the bottom-right corner of j . Then tl_x , tl_y , br_x , br_y , ctl_x , ctl_y , cbr_x and cbr_y should satisfy the following relationship:

$$\begin{cases} ctl_x = \frac{(n+1)tl_x + (n-1)br_x}{2n} \\ ctl_y = \frac{(n+1)tl_y + (n-1)br_y}{2n} \\ cbr_x = \frac{(n-1)tl_x + (n+1)br_x}{2n} \\ cbr_y = \frac{(n-1)tl_y + (n+1)br_y}{2n} \end{cases} \quad (1)$$

where n is odd that determines the scale of the central re-

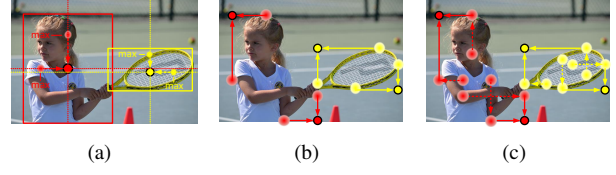


Figure 4: (a) Center pooling takes the maximum values in both horizontal and vertical directions. (b) Corner pooling only takes the maximum values in boundary directions. (c) Cascade corner pooling takes the maximum values in both boundary directions and internal directions of objects.

gion j . In this paper, n is set to be 3 and 5 for the scales of bounding boxes less and greater than 150, respectively. Figure 3 shows two central regions when $n = 3$ and $n = 5$, respectively. According to Equation (1), we can determine a scale-aware central region, then we check if the central region contains center keypoints.

3.3. Enriching Center and Corner Information

Center pooling. The geometric centers of objects do not necessarily convey very recognizable visual patterns (*e.g.*, the human head contains strong visual patterns, but the center keypoint is often in the middle of the human body). To address this issue, we propose center pooling to capture richer and more recognizable visual patterns. Figure 4(a) shows the principle of center pooling. The detailed process of center pooling is as follows: the backbone outputs a feature map, and to determine if a pixel in the feature map is a center keypoint, we need to find the maximum value in its both horizontal and vertical directions and add them together. By doing this, center pooling helps the better detection of center keypoints.

Cascade corner pooling. Corners are often outside the objects, which lacks local appearance features. CornerNet [20] uses corner pooling to address this issue. The principle of corner pooling is shown in Figure 4(b). Corner pooling aims to find the maximum values on the boundary directions so as to determine corners. However, it makes corners sensitive to the edges. To address this problem, we need to let corners “see” the visual patterns of objects. The principle of cascade corner pooling is presented in Figure 4(c). It first looks along a boundary to find a boundary maximum value, then looks inside along the location of the boundary maximum value² to find an internal maximum value, and finally, add the two maximum values together. By doing this, the corners obtain both the the boundary information and the visual patterns of objects.

Both the center pooling and the cascade corner pooling can be easily achieved by combining the corner pooling [20]

²For the topmost, leftmost, bottommost and rightmost boundary, look vertically towards the bottom, horizontally towards the right, vertically towards the top and horizontally towards the left, respectively.

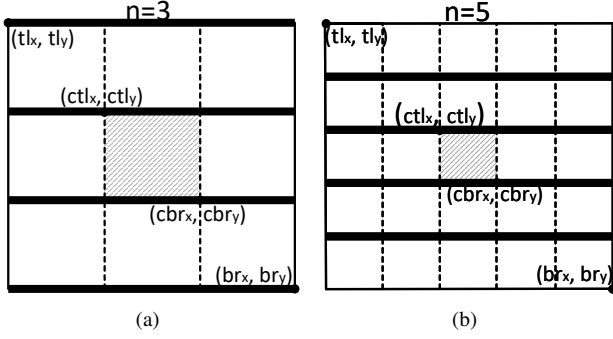


图3: (a) 当 $n = 3$ 时的中心区域。(b) 当 $n = 5$ 时的中心区域。实线矩形表示预测边界框，阴影区域表示中心区域。

为每个边界框定义一个中心区域，并检查该中心区域是否包含中心关键点。需要注意的是，所检查中心关键点的类别标签应与边界框的类别相同；(4) 如果在中心区域检测到中心关键点，我们将保留该边界框。边界框的分数将被替换为三个点的平均分数，即左上角、右下角和中心关键点的分数。如果在其中心区域未检测到中心关键点，则该边界框将被移除。

边界框中心区域的大小会影响检测结果。例如，较小的中心区域会导致小边界框的召回率较低，而较大的中心区域则会导致大边界框的精确度较低。因此，我们提出了一种尺度感知的中心区域，以自适应地拟合边界框的大小。尺度感知的中心区域倾向于为小边界框生成相对较大的中心区域，而为大边界框生成相对较小的中心区域。假设我们需要判断一个边界框 i 是否需要保留。令 tl_x 和 tl_y 表示 i 左上角的坐标， br_x 和 br_y 表示 i 右下角的坐标。定义一个中心区域 j 。令 ctl_x 和 ctl_y 表示 j 左上角的坐标， cbr_x 和 cbr_y 表示 j 右下角的坐标。那么 tl_x 、 tl_y 、 br_x 、 br_y 、 ctl_x 、 ctl_y 、 cbr_x 和 cbr_y 应满足以下关系：

$$\begin{cases} ctl_x = \frac{(n+1)tl_x + (n-1)br_x}{2n} \\ ctl_y = \frac{(n+1)tl_y + (n-1)br_y}{2n} \\ cbr_x = \frac{(n-1)tl_x + (n+1)br_x}{2n} \\ cbr_y = \frac{(n-1)tl_y + (n+1)br_y}{2n} \end{cases} \quad (1)$$

其中 n 是决定中心重标度尺度的奇数

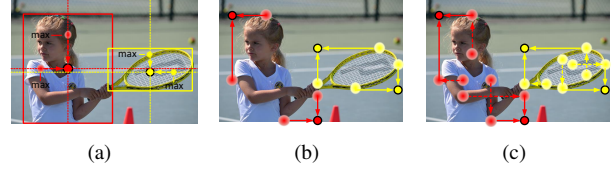


图4: (a) 中心池化同时提取水平和垂直方向的最大值。(b) 角点池化仅提取边界方向的最大值。(c) 级联角点池化同时提取物体边界方向与内部方向的最大值。

gion j 。在本文中，对于边界框尺度小于和大于150的情况， n 分别设为3和5。图3分别展示了当 $n = 3$ 和 $n = 5$ 时的两个中心区域。根据公式(1)，我们可以确定一个尺度感知的中心区域，然后检查该中心区域是否包含中心关键点。

3.3. 丰富中心与角点信息

中心池化。物体的几何中心并不总是传达非常可识别的视觉模式（e.g. 例如，人类头部包含强烈的视觉模式，但中心关键点通常位于人体的中部）。为了解决这个问题，我们提出了中心池化来捕捉更丰富、更易识别的视觉模式。图4(a)展示了中心池化的原理。中心池化的详细过程如下：骨干网络输出一个特征图，为了确定特征图中的某个像素是否为中心关键点，我们需要在其水平和垂直方向上找到最大值，并将它们相加。通过这种方式，中心池化有助于更好地检测中心关键点。

级联角点池化。角点通常位于物体外部，缺乏局部外观特征。CornerNet [20] 使用角点池化来解决这一问题。角点池化的原理如图4(b)所示，其通过沿边界方向寻找最大值来确定角点位置。但这种方法会使角点对边缘信息过于敏感。为解决此问题，我们需要让角点“感知”物体的视觉模式。级联角点池化的原理如图4(c)所示：它首先沿边界方向寻找边界最大值，随后沿该边界最大值²的位置向物体内部寻找内部最大值，最终将这两个最大值相加。通过这种方式，角点同时获得了边界信息和物体的视觉模式。

中心池化和级联角点池化都可以通过结合角池化[20]轻松实现。

²For the topmost, leftmost, bottommost and rightmost boundary, look vertically towards the bottom, horizontally towards the right, vertically towards the top and horizontally towards the left, respectively.

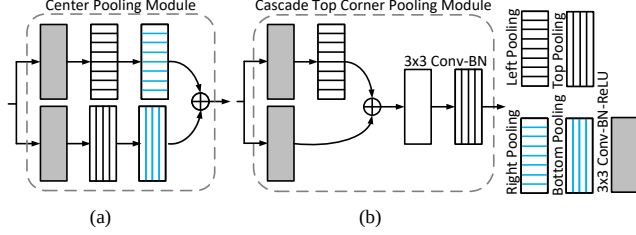


Figure 5: The structures of the center pooling module (a) and the cascade top corner pooling module (b). We achieve center pooling and the cascade corner pooling by combining the corner pooling at different directions.

at different directions. Figure 5(a) shows the structure of the center pooling module. To take a maximum value in a direction, *e.g.*, the horizontal direction, we only need to connect the left pooling and the right pooling in series. Figure 5(b) shows the structure of a cascade top corner pooling module. Compared with the top corner pooling in CornerNet [20], we add a left corner pooling before the top corner pooling.

3.4. Training and Inference

Training. Our method is implemented in Pytorch [30] and the network is trained from scratch. The resolution of the input image is 511×511 , leading to heatmaps of size 128×128 . We use the data augmentation strategy presented in [20] to train a robust model. Adam [18] is used to optimize the training loss:

$$L = L_{\text{det}}^{\text{co}} + L_{\text{det}}^{\text{ce}} + \alpha L_{\text{pull}}^{\text{co}} + \beta L_{\text{push}}^{\text{co}} + \gamma (L_{\text{off}}^{\text{co}} + L_{\text{off}}^{\text{ce}}), \quad (2)$$

where $L_{\text{det}}^{\text{co}}$ and $L_{\text{det}}^{\text{ce}}$ denote the focal losses, which are used to train the network to detect corners and center keypoints, respectively. $L_{\text{pull}}^{\text{co}}$ is a “pull” loss for corners, which is used to minimize the distance of the embedding vectors that belongs to the same objects. $L_{\text{push}}^{\text{co}}$ is a “push” loss for corners, which is used to maximize the distance of the embedding vectors that belongs to different objects. $L_{\text{off}}^{\text{co}}$ and $L_{\text{off}}^{\text{ce}}$ are ℓ_1 -losses [10], which are used to train the network to predict the offsets of corners and center keypoints, respectively. α , β and γ denote the weights for corresponding losses, which are set to 0.1, 0.1 and 1, respectively. L_{det} , L_{pull} , L_{push} and L_{off} are all defined in the CornerNet, we suggest to refer to [20] for details. We train the CenterNet on 8 Tesla V100 (32GB) GPUs and use a batch size of 48. The maximum number of iterations is 480K. We use a learning rate of 2.5×10^{-4} for the first 450K iterations and then continue training 30K iterations with a rate of 2.5×10^{-5} .

Inference. Following [20], for the single-scale testing, we input both the original and horizontally flipped images with the original resolutions into the network. While for the multi-scale testing, we input both the original and horizontally flipped images with the resolutions of 0.6, 1, 1.2, 1.5 and 1.8. We select top 70 center keypoints, top 70 top-left

corners and top 70 bottom-right corners from the heatmaps to detect the bounding boxes. We flip the bounding boxes detected in the horizontally flipped images and mix them into the original bounding boxes. Soft-nms [2] is used to remove the redundant bounding boxes. We finally select top 100 bounding boxes according to their scores as the final detection results.

4. Experiments

4.1. Dataset, Metrics and Baseline

We evaluate our method on the MS-COCO dataset [25]. It contains 80 categories and more than 1.5 million object instances. The large number of small objects makes it a very challenging dataset. We use the ‘trainval35k’ set [15] (*i.e.*, 80K training images and 35K validation images) for training and test the results on the test-dev set. We use another 5K images in the validation set to perform ablation studies and visualization experiments.

MS-COCO dataset [25] uses AP and AR metrics to characterize the performance of a detector. AP represents the average precision rate, which is computed over ten different IoU thresholds (*i.e.*, 0.5 : 0.05 : 0.95) and all categories. It is considered the single most important metric on the MS-COCO dataset. AR represents the maximum recall rate, which is computed over a fixed number of detections (*i.e.*, 1, 10 and 100) per image and averaged over all categories and the ten different IoU thresholds. Additionally, AP and AR can be used to evaluate the performance under different object scales, including small objects ($\text{area} < 32^2$), medium objects ($32^2 < \text{area} < 96^2$) and large objects ($\text{area} > 96^2$).

Our direct baseline is CornerNet [20]. Following it, we use the stacked hourglass network [29] with 52 and 104 layers as the backbone – the latter has two hourglass modules while the former has only one. All modifications on the hourglass architecture, made by [20], are preserved.

4.2. Comparisons with State-of-the-art Detectors

Table 2 shows the comparison with the state-of-the-art detectors on the MS-COCO test-dev set.

Compared with the baseline CornerNet [20], the proposed CenterNet achieves a remarkable improvement. For example, CenterNet511-52 (means that the resolution of input images is 511×511 and the backbone is Hourglass-52) reports a single-scale testing AP of 41.6%, an improvement of 3.8% over 37.8%, and a multi-scale testing AP of 43.5%, an improvement of 4.1% over 39.4%, achieved by CornerNet under the same setting. When using the deeper backbone (*i.e.*, Hourglass-104), the AP improvement over CornerNet are 4.4% (from 40.5% to 44.9%) and 4.9% (from 42.1% to 47.0%) under the single-scale and multi-scale testing, respectively. These results firmly demonstrate the effectiveness of CenterNet.

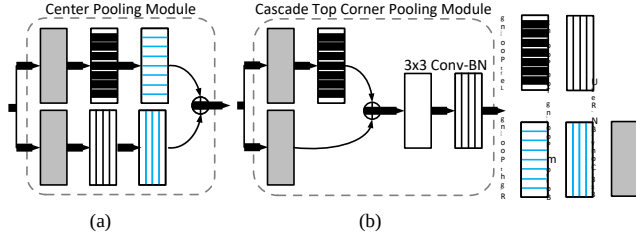


图5: 中心池化模块 (a) 和级联顶角池化模块 (b) 的结构。我们通过组合不同方向的角池化, 实现了中心池化与级联角池化。

在不同方向上。图5(a)展示了中心池化模块的结构。为了在特定方向(例如水平方向 $\{v^*\}$)获取最大值, 我们只需将左池化与右池化串联连接。图5(b)展示了级联顶部角点池化模块的结构。与CornerNet[20]中的顶部角点池化相比, 我们在顶部角点池化之前增加了一个左侧角点池化。

3.4. 训练与推理

训练。我们的方法在Pytorch [30] 中实现, 网络从头开始训练。输入图像的分辨率为 511×511 , 生成的热图尺寸为 128×128 。我们采用文献[20]提出的数据增强策略来训练鲁棒模型, 并使用Adam [18]优化器来最小化训练损失:

$$L = L_{\text{det}}^{\text{co}} + L_{\text{det}}^{\text{ce}} + \alpha L_{\text{pull}}^{\text{co}} + \beta L_{\text{push}}^{\text{co}} + \gamma (L_{\text{off}}^{\text{co}} + L_{\text{off}}^{\text{ce}}), \quad (2)$$

其中 $L_{\text{det}}^{\text{co}}$ 和 $L_{\text{det}}^{\text{ce}}$ 表示焦点损失, 分别用于训练网络检测角点和中心关键点。 $L_{\text{pull}}^{\text{co}}$ 是角点的“吸引”损失, 用于最小化属于同一目标的嵌入向量之间的距离。 $L_{\text{push}}^{\text{co}}$ 是角点的“排斥”损失, 用于最大化属于不同目标的嵌入向量之间的距离。 $L_{\text{off}}^{\text{co}}$ 和 $L_{\text{off}}^{\text{ce}}$ 是 ℓ_1 损失[10], 分别用于训练网络预测角点和中心关键点的偏移量。 α 、 β 和 γ 表示对应损失的权重, 分别设置为0.1、0.1和1。 L_{det} 、 L_{pull} 、 L_{push} 和 L_{off} 均在CornerNet中定义, 建议参阅[20]了解细节。我们在8块Tesla V100 (32GB) GPU上训练CenterNet, 批次大小设为48。最大迭代次数为480K。前450K次迭代使用 2.5×10^{-4} 的学习率, 随后以 2.5×10^{-5} 的学习率继续训练30K次迭代。

推理。遵循[20]的方法, 在单尺度测试中, 我们将原始分辨率图像及其水平翻转图像输入网络。而在多尺度测试中, 我们则输入原始分辨率图像及其水平翻转图像, 分辨率分别为0.6, 1, 1.2, 1.5和1.8。我们选取置信度最高的70个中心关键点、70个左上角关键点

我们从热图中选取左上角的前70个角点和右下角的前70个角点来检测边界框。对于水平翻转图像中检测到的边界框, 我们会将其翻转后与原始边界框合并。采用Soft-NMS[2]方法去除冗余边界框。最后, 我们根据得分选取前100个边界框作为最终的检测结果。

4. 实验

4.1. 数据集、指标与基线

我们在MS-COCO数据集[25]上评估了我们的方法。该数据集包含80个类别和超过1.5百万个物体实例。大量的小物体使其成为一个极具挑战性的数据集。我们使用“trainval35k”集[15] (*i.e.*, 即8万张训练图像和3.5万张验证图像) 进行训练, 并在test-dev集上测试结果。我们使用验证集中的另外5千张图像进行消融研究和可视化实验。

MS-COCO数据集[25]使用AP和AR指标来表征检测器的性能。AP代表平均精度率, 其计算基于十个不同的IoU阈值 (*i.e.*, 0.5 : 0.05 : 0.95) 和所有类别。它被认为是MS-COCO数据集中最重要的单一指标。AR代表最大召回率, 其计算基于每张图像固定数量的检测结果 (*i.e.*, 1, 10和100), 并在所有类别和十个不同IoU阈值上取平均值。此外, AP和AR可用于评估不同物体尺度下的性能, 包括小物体 (面积 $< 32^2$)、中等物体 ($32^2 < \text{面积} < 96^2$) 和大物体 (面积 $> 96^2$)。

我们的直接基线是CornerNet [20]。遵循其方法, 我们使用具有52层和104层的堆叠沙漏网络[29]作为主干网络——后者包含两个沙漏模块, 而前者仅有一个。[20]对沙漏架构所做的所有修改均被保留。

4.2. 与最先进检测器的比较

表2展示了在MS-COCO测试开发集上与最先进检测器的比较。

与基线CornerNet [20]相比, 所提出的CenterNet取得了显著提升。例如, CenterNet511-52 (表示输入图像分辨率为 511×511 , 主干网络为Hourglass-52) 在单尺度测试中达到41.6%的AP, 较CornerNet在相同设置下取得的37.8%提升了3.8%; 多尺度测试AP为43.5%, 较CornerNet的39.4%提升了4.1%。当使用更深的主干网络 (即Hourglass-104) 时, 在单尺度与多尺度测试中, AP分别较CornerNet提升了4.4% (从40.5%提升至44.9%) 和4.9% (从42.1%提升至47.0%)。这些结果有力证明了CenterNet的有效性。

Method	Backbone	Train input	Test input	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L	AR ₁	AR ₁₀	AR ₁₀₀	AR _S	AR _M	AR _L
Two-stage:															
DeNet [40]	ResNet-101 [14]	512×512	512×512	33.8	53.4	36.1	12.3	36.1	50.8	29.6	42.6	43.5	19.2	46.9	64.3
CoupleNet [47]	ResNet-101	ori.	ori.	34.4	54.8	37.2	13.4	38.1	50.8	30.0	45.0	46.4	20.7	53.1	68.5
Faster R-CNN by G-RMI [16]	Inception-ResNet-v2 [39]	~1000×600	~1000×600	34.7	55.5	36.7	13.5	38.1	52.0	-	-	-	-	-	-
Faster R-CNN +++ [14]	ResNet-101	~1000×600	~1000×600	34.9	55.7	37.4	15.6	38.7	50.9	-	-	-	-	-	-
Faster R-CNN w/ FPN [23]	ResNet-101	~1000×600	~1000×600	36.2	59.1	39.0	18.2	39.0	48.2	-	-	-	-	-	-
Faster R-CNN w/ TDM [37]	Inception-ResNet-v2	-	-	36.8	57.7	39.2	16.2	39.8	52.1	31.6	49.3	51.9	28.1	56.6	71.1
D-FCN [7]	Aligned-Inception-ResNet	~1000×600	~1000×600	37.5	58.0	-	19.4	40.1	52.5	-	-	-	-	-	-
Regionlets [43]	ResNet-101	~1000×600	~1000×600	39.3	59.8	-	21.7	43.7	50.9	-	-	-	-	-	-
Mask R-CNN [12]	ResNeXt-101	~1300×800	~1300×800	39.8	62.3	43.4	22.1	43.2	51.2	-	-	-	-	-	-
Soft-NMS [2]	Aligned-Inception-ResNet	~1300×800	~1300×800	40.9	62.8	-	23.3	43.6	53.3	-	-	-	-	-	-
Fitness R-CNN [41]	ResNet-101	512×512	1024×1024	41.8	60.9	44.9	21.5	45.0	57.5	-	-	-	-	-	-
Cascade R-CNN [4]	ResNet-101	-	-	42.8	62.1	46.3	23.7	45.5	55.2	-	-	-	-	-	-
Grid R-CNN w/ FPN [28]	ResNeXt-101	~1300×800	~1300×800	43.2	63.0	46.6	25.1	46.5	55.2	-	-	-	-	-	-
D-RFCN + SNIP (multi-scale) [38]	DPN-98 [5]	~2000×1200	~2000×1200	45.7	67.3	51.1	29.3	48.8	57.1	-	-	-	-	-	-
PANet (multi-scale) [26]	ResNeXt-101	~1400×840	~1400×840	47.4	67.2	51.8	30.1	51.7	60.0	-	-	-	-	-	-
One-stage:															
YOLOv2 [32]	DarkNet-19	544×544	544×544	21.6	44.0	19.2	5.0	22.4	35.5	20.7	31.6	33.3	9.8	36.5	54.4
DSOD300 [34]	DS/64-192-48-1	300×300	300×300	29.3	47.3	30.6	9.4	31.5	47.0	27.3	40.7	43.0	16.7	47.1	65.0
GRP-DSOD320 [35]	DS/64-192-48-1	320×320	320×320	30.0	47.9	31.8	10.9	33.6	46.3	28.0	42.1	44.5	18.8	49.1	65.0
SSD513 [27]	ResNet-101	513×513	513×513	31.2	50.4	33.3	10.2	34.5	49.8	28.3	42.1	44.4	17.6	49.2	65.8
DSSD513 [8]	ResNet-101	513×513	513×513	33.2	53.3	35.2	13.0	35.4	51.1	28.9	43.5	46.2	21.8	49.1	66.4
RefineDet512 (single-scale) [45]	ResNet-101	512×512	512×512	36.4	57.5	39.5	16.6	39.9	51.4	-	-	-	-	-	-
CornerNet511 (single-scale) [20]	Hourglass-52	511×511	ori.	37.8	53.7	40.1	17.0	39.0	50.5	33.9	52.3	57.0	35.0	59.3	74.7
RetinaNet800 [24]	ResNet-101	800×800	800×800	39.1	59.1	42.3	21.8	42.7	50.2	-	-	-	-	-	-
CornerNet511 (multi-scale) [20]	Hourglass-52	511×511	≤1.5×	39.4	54.9	42.3	18.9	41.2	52.7	35.0	53.5	57.7	36.1	60.1	75.1
CornerNet511 (single-scale) [20]	Hourglass-104	511×511	ori.	40.5	56.5	43.1	19.4	42.7	53.9	35.3	54.3	59.1	37.4	61.9	76.9
RefineDet512 (multi-scale) [45]	ResNet-101	512×512	≤2.25×	41.8	62.9	45.7	25.6	45.1	54.1	-	-	-	-	-	-
CornerNet511 (multi-scale) [20]	Hourglass-104	511×511	≤1.5×	42.1	57.8	45.3	20.8	44.8	56.7	36.4	55.7	60.0	38.5	62.7	77.4
CenterNet511 (single-scale)	Hourglass-52	511×511	ori.	41.6	59.4	44.2	22.5	43.1	54.1	34.8	55.7	60.1	38.6	63.3	76.9
CenterNet511 (single-scale)	Hourglass-104	511×511	ori.	44.9	62.4	48.1	25.6	47.4	57.4	36.1	58.4	63.3	41.3	67.1	80.2
CenterNet511 (multi-scale)	Hourglass-52	511×511	≤1.8×	43.5	61.3	46.7	25.3	45.3	55.0	36.0	57.2	61.3	41.4	64.0	76.3
CenterNet511 (multi-scale)	Hourglass-104	511×511	≤1.8×	47.0	64.5	50.7	28.9	49.9	58.9	37.5	60.3	64.8	45.1	68.3	79.7

Table 2: Performance comparison (%) with the state-of-the-art methods on the MS-COCO test-dev dataset. CenterNet outperforms all existing one-stage detectors by a large margin and ranks among the top of state-of-the-art two-stage detectors.

Meanwhile, it can be seen that the most contribution comes from the small objects. For instance, CenterNet511-52 improves the AP for small objects by 5.5% (single-scale) and by 6.4% (multi-scale). As for the backbone Hourglass-104, the improvements are 6.2% (single-scale) and by 8.1% (multi-scale), respectively. The benefit stems from the center information modeled by the center keypoints: the smaller the scale of an incorrect bounding box is, the lower probability a center keypoint can be detected in its central region. Figure 6(a) and Figure 6(b) show some qualitative comparisons, which demonstrate the effectiveness of CenterNet in reducing small incorrect bounding boxes.

CenterNet also leads to a large improvement for reducing medium and large incorrect bounding boxes. As Table 2 shows, CenterNet511-104 improves the single-scale testing AP by 4.7% (from 42.7% to 47.4%) and 3.5% (from 53.9% to 57.4%), respectively. Figure 6(c) and Figure 6(d) show some qualitative comparisons for reducing medium and large incorrect bounding boxes. It is worth noting that the AR is also significantly improved, with the best performance achieved with multi-scale testing. This is because our approach removes lots of incorrect bounding boxes, which is equivalent to improving the confidence of those bounding boxes with accurate locations but lower scores.

When comparing other one-stage approaches, CenterNet511-52 reports 41.6% single-scale testing AP. This achievement is already better than those using deeper models (*e.g.*, RetinaNet800 [24] and RefineDet [45]). The best performance of CenterNet is AP **47.0%**, dramatically

surpassing all the published one-stage approaches to our best knowledge.

At last, one can observe that the performance of CenterNet is also competitive with the two-stage approaches, *e.g.*, the single-scale testing AP of CenterNet511-52 is comparable to the two-stage approach Fitness R-CNN [41] (41.6% vs. 41.8%) and that of CenterNet511-104 is comparable to D-RFCN + SNIP [38] (44.9% vs. 45.7%), respectively. Nevertheless, it should be mentioned that two-stage approaches usually use larger resolution input images (*e.g.*, $\sim 1000 \times 600$), which significantly improves the detection accuracy especially for small objects. The multi-scale testing AP **47.0%** achieved by CenterNet511-104 closely matches the state-of-the-art AP 47.4%, achieved by the two-stage detector PANet [26]. We present some qualitative detection results in Figure 7.

4.3. Incorrect Bounding Box Reduction

The AP [25] metric reflects how many high quality object bounding boxes (usually $\text{IoU} \geq 0.5$) a network can predict, but cannot directly reflect how many incorrect object bounding boxes (usually $\text{IoU} \ll 0.5$) a network generates. The FD rate is a suitable metric, which reflects the proportion of the incorrect bounding boxes. Table 3 shows the FD rates for CornerNet and CenterNet. CornerNet generates many incorrect bounding boxes even at $\text{IoU} = 0.05$ threshold, *i.e.*, CornerNet511-52 and CornerNet511-104 obtain 35.2% and 32.7% FD rate, respectively. On the other hand, CornerNet generates more small incorrect bounding

Method	Backbone	Train input	Test input	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L	AR ₁	AR ₁₀	AR ₁₀₀	AR _S	AR _M	AR _L
Two-stage:															
DeNet [40]	ResNet-101 [14]	512×512	512×512	33.8	53.4	36.1	12.3	36.1	50.8	29.6	42.6	43.5	19.2	46.9	64.3
CoupleNet [47]	ResNet-101	ori.	ori.	34.4	54.8	37.2	13.4	38.1	50.8	30.0	45.0	46.4	20.7	53.1	68.5
Faster R-CNN by G-RMI [16]	Inception-ResNet-v2 [39]	~ 1000×600	~ 1000×600	34.7	55.5	36.7	13.5	38.1	52.0	-	-	-	-	-	-
Faster R-CNN +++ [14]	ResNet-101	~ 1000×600	~ 1000×600	34.9	55.7	37.4	15.6	38.7	50.9	-	-	-	-	-	-
Faster R-CNN w/ FPN [23]	ResNet-101	~ 1000×600	~ 1000×600	36.2	59.1	39.0	18.2	39.0	48.2	-	-	-	-	-	-
Faster R-CNN w/ TDM [37]	Inception-ResNet-v2	-	-	36.8	57.7	39.2	16.2	39.8	52.1	31.6	49.3	51.9	28.1	56.6	71.1
D-FCN [7]	Aligned-Inception-ResNet	~ 1000×600	~ 1000×600	37.5	58.0	-	19.4	40.1	52.5	-	-	-	-	-	-
Regionlets [43]	ResNet-101	~ 1000×600	~ 1000×600	39.3	59.8	-	21.7	43.7	50.9	-	-	-	-	-	-
Mask R-CNN [12]	ResNeXt-101	~ 1300×800	~ 1300×800	39.8	62.3	43.4	22.1	43.2	51.2	-	-	-	-	-	-
Soft-NMS [2]	Aligned-Inception-ResNet	~ 1300×800	~ 1300×800	40.9	62.8	-	23.3	43.6	53.3	-	-	-	-	-	-
Fitness R-CNN [41]	ResNet-101	512×512	1024×1024	41.8	60.9	44.9	21.5	45.0	57.5	-	-	-	-	-	-
Cascade R-CNN [4]	ResNet-101	-	-	42.8	62.1	46.3	23.7	45.5	55.2	-	-	-	-	-	-
Grid R-CNN w/ FPN [28]	ResNeXt-101	~ 1300×800	~ 1300×800	43.2	63.0	46.6	25.1	46.5	55.2	-	-	-	-	-	-
D-RFCN + SNIP (multi-scale) [38]	DPN-98 [5]	~ 2000×1200	~ 2000×1200	45.7	67.3	51.1	29.3	48.8	57.1	-	-	-	-	-	-
PANet (multi-scale) [26]	ResNeXt-101	~ 1400×840	~ 1400×840	47.4	67.2	51.8	30.1	51.7	60.0	-	-	-	-	-	-
One-stage:															
YOLOv2 [32]	DarkNet-19	544×544	544×544	21.6	44.0	19.2	5.0	22.4	35.5	20.7	31.6	33.3	9.8	36.5	54.4
DSOD300 [34]	DS/64-192-48-1	300×300	300×300	29.3	47.3	30.6	9.4	31.5	47.0	27.3	40.7	43.0	16.7	47.1	65.0
GRP-DSOD320 [35]	DS/64-192-48-1	320×320	320×320	30.0	47.9	31.8	10.9	33.6	46.3	28.0	42.1	44.5	18.8	49.1	65.0
SSD513 [27]	ResNet-101	513×513	513×513	31.2	50.4	33.3	10.2	34.5	49.8	28.3	42.1	44.4	17.6	49.2	65.8
DSSD513 [8]	ResNet-101	513×513	513×513	33.2	53.3	35.2	13.0	35.4	51.1	28.9	43.5	46.2	21.8	49.1	66.4
RefineDet512 (single-scale) [45]	ResNet-101	512×512	512×512	36.4	57.5	39.5	16.6	39.9	51.4	-	-	-	-	-	-
CornerNet511 (single-scale) [20]	Hourglass-52	511×511	ori.	37.8	53.7	40.1	17.0	39.0	50.5	33.9	52.3	57.0	35.0	59.3	74.7
RetinaNet800 [24]	ResNet-101	800×800	800×800	39.1	59.1	42.3	21.8	42.7	50.2	-	-	-	-	-	-
CornerNet511 (multi-scale) [20]	Hourglass-52	511×511	≤1.5×	39.4	54.9	42.3	18.9	41.2	52.7	35.0	53.5	57.7	36.1	60.1	75.1
CornerNet511 (single-scale) [20]	Hourglass-104	511×511	ori.	40.5	56.5	43.1	19.4	42.7	53.9	35.3	54.3	59.1	37.4	61.9	76.9
RefineDet512 (multi-scale) [45]	ResNet-101	512×512	≤2.25×	41.8	62.9	45.7	25.6	45.1	54.1	-	-	-	-	-	-
CornerNet511 (multi-scale) [20]	Hourglass-104	511×511	≤1.5×	42.1	57.8	45.3	20.8	44.8	56.7	36.4	55.7	60.0	38.5	62.7	77.4
CenterNet511 (single-scale)	Hourglass-52	511×511	ori.	41.6	59.4	44.2	22.5	43.1	54.1	34.8	55.7	60.1	38.6	63.3	76.9
CenterNet511 (single-scale)	Hourglass-104	511×511	ori.	44.9	62.4	48.1	25.6	47.4	57.4	36.1	58.4	63.3	41.3	67.1	80.2
CenterNet511 (multi-scale)	Hourglass-52	511×511	≤1.8×	43.5	61.3	46.7	25.3	45.3	55.0	36.0	57.2	61.3	41.4	64.0	76.3
CenterNet511 (multi-scale)	Hourglass-104	511×511	≤1.8×	47.0	64.5	50.7	28.9	49.9	58.9	37.5	60.3	64.8	45.1	68.3	79.7

表2：在MS-COCO test-dev数据集上与最先进方法的性能对比（%）。CenterNet以显著优势超越所有现有的一阶段检测器，并位列最先进二阶段检测器的前列。

同时可以看出，最主要的贡献来自小物体。例如，CenterNet511-52将小物体的AP提升了5.5%（单尺度）和6.4%（多尺度）。而对于骨干网络Hourglass-104，提升分别为6.2%（单尺度）和8.1%（多尺度）。这种优势源于中心关键点建模的中心信息：错误边界框的尺度越小，其中心区域检测到中心关键点的概率就越低。图6(a)和图6(b)展示了一些定性比较，证明了CenterNet在减少小尺寸错误边界框方面的有效性。

CenterNet在减少中大型错误边界框方面也带来了显著改进。如表2所示，CenterNet511-104将单尺度测试AP分别提升了4.7%（从42.7%提升至47.4%）和3.5%（从53.9%提升至57.4%）。图6(c)与图6(d)展示了减少中大型错误边界框的定性对比结果。值得注意的是，AR指标也得到显著提升，其中多尺度测试取得了最佳性能。这是因为我们的方法消除了大量错误边界框，这相当于提升了那些位置准确但得分较低的边界框的置信度。

在与其他单阶段方法比较时，CenterNet511-52报告了41.6%的单尺度测试AP。这一成果已经优于使用更深层模型的方法（e.g., 如RetinaNet800 [24]和RefineDet [45]）。CenterNet的最佳性能为AP 47.0%，显著

据我们所知，超越了所有已发表的一阶段方法。

最后，可以观察到CenterNet的性能也与两阶段方法具有竞争力，e.g., CenterNet511-52的单尺度测试AP与两阶段方法Fitness R-CNN [41]相当（41.6% vs. 41.8%），而CenterNet511-104则与D-RFCN + SNIP [38]相当（44.9% vs. 45.7%）。然而，需要指出的是，两阶段方法通常使用更高分辨率的输入图像（e.g., ~ 1000 × 600），这显著提高了检测精度，特别是对于小物体。CenterNet511-104实现的多尺度测试AP 47.0%与最先进的AP 47.4%非常接近，后者由两阶段检测器PANet [26]实现。我们在图7中展示了一些定性检测结果。

4.3. 不正确的边界框缩减

AP [25] 指标反映了网络能够预测出多少个高质量的目标边界框（通常IoU ≥ 0.5），但无法直接反映网络会产生多少个错误的目标边界框（通常IoU < 0.5）。FD率是一个合适的指标，它反映了错误边界框的比例。表3展示了CornerNet和CenterNet的FD率。即使在IoU = 0.05的阈值下，CornerNet也会产生许多错误的边界框，i.e.例如，CornerNet511-52和CornerNet511-104分别获得了35.2%和32.7%的FD率。另一方面，CornerNet会产生更多小的错误边界

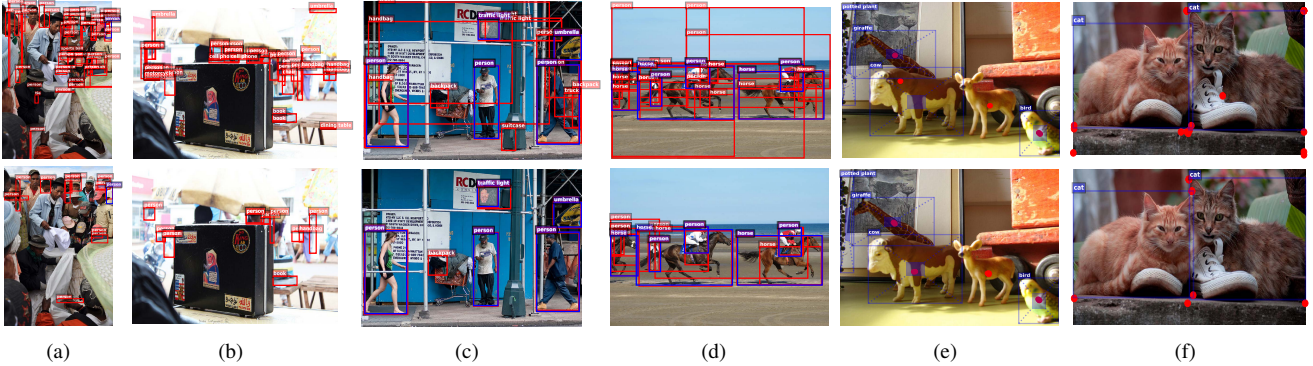


Figure 6: (a) and (b) show the small incorrect bounding boxes are significantly reduced by modeling center information. (c) and (d) show that the center information works for reducing medium and large incorrect bounding boxes. (e) shows the results of detecting the center keypoints without/with the center pooling. (f) shows the results of detecting the corners with corner pooling and cascade corner pooling, respectively. The blue boxes above denote the ground-truth. The red boxes and dots denote the predicted bounding boxes and keypoints, respectively.



Figure 7: Some qualitative detection results on the MS-COCO validation dataset. Only detections with scores higher than 0.5 are shown.

Method	FD	FD ₅	FD ₂₅	FD ₅₀	FD _S	FD _M	FD _L
CornerNet511-52	40.4	35.2	39.4	46.7	62.5	36.9	28.0
CenterNet511-52	35.1	30.7	34.2	40.8	53.0	31.3	24.4
CornerNet511-104	37.8	32.7	36.8	43.8	60.3	33.2	25.1
CenterNet511-104	32.4	28.2	31.6	37.5	50.7	27.1	23.0

Table 3: Comparison of false discovery rates (%) of CornerNet and CenterNet on the MS-COCO validation dataset. The results suggest CenterNet avoids a large number of incorrect bounding boxes, especially for small incorrect bounding boxes.

boxes than medium and large incorrect bounding boxes, which reports 62.5% for CornerNet511-52 and 60.3% for CornerNet511-104, respectively. Our CenterNet decreases the FD rates at all criteria via exploring central regions. For instance, CenterNet511-52 and CenterNet511-104 decrease FD₅ by both 4.5%. In addition, the FD rates for small bounding boxes decrease the most, which are 9.5% by CenterNet511-52 and 9.6% by CenterNet511-104, respectively. This is also the reason why the AP improvement for small objects is more prominent.

4.4. Inference Speed

The proposed CenterNet explores the visual patterns within each proposed region with minimal costs. To ensure a fair comparison, we test the inference speed of both CornerNet [20] and CenterNet on a NVIDIA Tesla P100 GPU. We obtain that the average inference time of CornerNet511-

104 is 300ms per image and that of CenterNet511-104 is 340ms. Meanwhile, using the Hourglass-52 backbone can speed up the inference speed. Our CenterNet511-52 takes an average of 270ms to process per image, which is faster and more accurate than CornerNet511-104.

4.5. Ablation Study

Our work has contributed three components, including central region exploration, center pooling and cascade corner pooling. To analyze the contribution of each individual component, an ablation study is given here. The baseline is CornerNet511-52 [20]. We add the three components to the baseline one by one and follow the default parameter setting detailed in Section 4.1. The results are given in Table 4.

Central region exploration. To understand the importance of the central region exploration (see CRE in the table), we add a center heatmap branch to the baseline and use a triplet of keypoints to detect bounding boxes. For the center keypoint detection, we only use conventional convolutions. As presented in the third row in Table 4, we improve the AP by 2.3% (from 37.6% to 39.9%). However, we find that the improvement for the small objects (that is 4.6%) is more significant than that for other object scales. The improvement for large objects is almost negligible (from 52.2% to 52.3%). This is not surprising because, from a probabilistic point of view, the center keypoint for a small object is easier

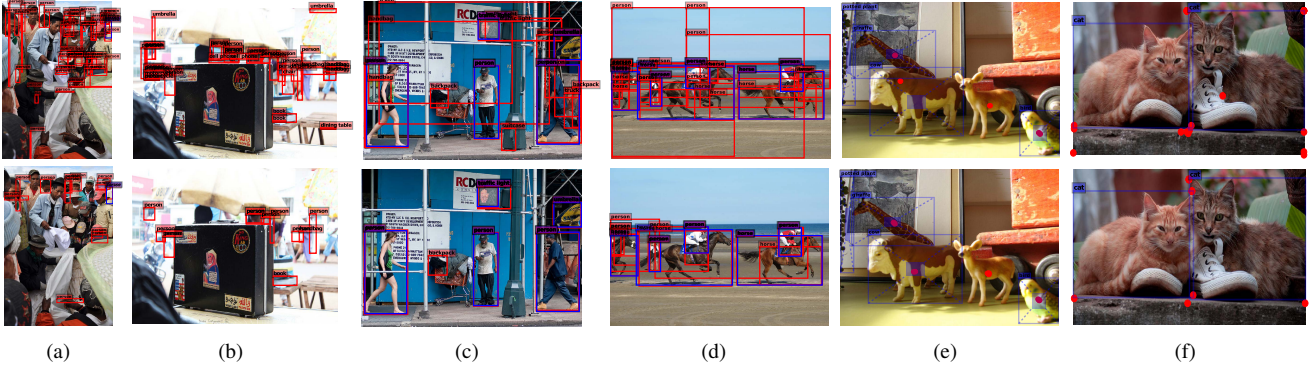


图6: (a)和(b)显示, 通过建模中心信息, 小尺寸错误边界框显著减少。(c)和(d)表明中心信息对减少中、大尺寸错误边界框同样有效。(e)展示了不使用/使用中心池化时中心关键点检测的结果。(f)分别展示了使用角点池化与级联角点池化检测角点的结果。上方蓝色框表示真实标注框, 红色框与红点分别表示预测的边界框与关键点。



图7: 在MS-COCO验证数据集上的一些定性检测结果。仅显示得分高于0.5的检测结果。

Method	FD	FD ₅	FD ₂₅	FD ₅₀	FD _S	FD _M	FD _L
CornerNet511-52	40.4	35.2	39.4	46.7	62.5	36.9	28.0
CenterNet511-52	35.1	30.7	34.2	40.8	53.0	31.3	24.4
CornerNet511-104	37.8	32.7	36.8	43.8	60.3	33.2	25.1
CenterNet511-104	32.4	28.2	31.6	37.5	50.7	27.1	23.0

表3: CornerNet与CenterNet在MS-COCO验证数据集上的误检率(%)对比。结果表明CenterNet避免了大量错误边界框, 尤其是针对小型错误边界框的情况。

与中型和大型错误边界框相比, 小型边界框的误检率更高, CornerNet511-52和CornerNet511-104分别报告了62.5%和60.3%的误检率。我们的CenterNet通过探索中心区域, 在所有标准下都降低了误检率。例如, CenterNet511-52和CenterNet511-104均将误检率降低了4.5%。此外, 小型边界框的误检率下降最为显著, CenterNet511-52和CenterNet511-104分别降低了9.5%和9.6%。这也是小物体AP提升更为突出的原因。

4.4. 推理速度

提出的CenterNet以最小成本探索每个提议区域内的视觉模式。为确保公平比较, 我们在NVIDIA Tesla P100 GPU上测试了CornerNet [20]和CenterNet的推理速度。测得CornerNet511-

104的每张图像处理时间为300毫秒, 而CenterNet511-104为340毫秒。同时, 使用Hourglass-52骨干网络可以加快推理速度。我们的CenterNet511-52平均每张图像处理时间为270毫秒, 比CornerNet511-104更快且更准确。

4.5. 消融研究

我们的工作贡献了三个组成部分, 包括中心区域探索、中心池化和级联角点池化。为了分析每个单独组件的贡献, 这里进行了一项消融研究。基线是CornerNet511-52 [20]。我们将这三个组件逐一添加到基线中, 并遵循第4.1节详述的默认参数设置。结果如表4所示。

中心区域探索。为了理解中心区域探索的重要性(见表格中的CRE), 我们在基线模型上增加了一个中心热图分支, 并使用关键点三元组来检测边界框。对于中心关键点检测, 我们仅使用常规卷积。如表4第三行所示, 我们将AP提升了2.3%(从37.6%提升至39.9%)。然而, 我们发现小目标(即4.6%)的提升比其他目标尺度更为显著。大目标的提升几乎可以忽略不计(从52.2%提升至52.3%)。这并不令人意外, 因为从概率角度来看, 小目标的中心关键点更容易

CRE	CTP	CCP	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L	AR ₁	AR ₁₀	AR ₁₀₀	AR _S	AR _M	AR _L
			37.6	53.3	40.0	18.5	39.6	52.2	33.7	52.2	56.7	37.2	60.0	74.0
		✓	38.3	54.2	40.5	18.6	40.5	52.2	34.0	53.0	57.9	36.6	60.8	75.8
✓			39.9	57.7	42.3	23.1	42.3	52.3	33.8	54.2	58.5	38.7	62.4	74.4
✓	✓		40.8	58.6	43.6	23.6	43.6	53.6	33.9	54.5	59.0	39.0	63.2	74.7
✓	✓	✓	41.3	59.2	43.9	23.6	43.8	55.8	34.5	55.0	59.2	39.1	63.5	75.1

Table 4: Ablation study on the major components of CenterNet511-52 on the MS-COCO validation dataset. The CRE denotes central region exploration, the CTP denotes center pooling, and the CCP denotes cascade corner pooling.

Method	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
CenterNet511-52 w/o GT	41.3	59.2	43.9	23.6	43.8	55.8
CenterNet511-52 w/ GT	56.5	78.3	61.4	39.1	60.3	70.3
CenterNet511-104 w/o GT	44.8	62.4	48.2	25.9	48.9	58.8
CenterNet511-104 w/ GT	58.1	78.4	63.9	40.4	63.0	72.1

Table 5: Error analysis of center keypoints via using ground-truth. we replace the predicted center keypoints with the ground-truth values, the results suggest there is still room for improvement in detecting center keypoints.

to be located than that of a large object.

Center pooling. To demonstrate the effectiveness of proposed center pooling, we then add the center pooling module to the network (see CTP in the table). The fourth row in Table 4 shows that center pooling improves the AP by 0.9% (from 39.9% to 40.8%). Notably, with the help of center pooling, we improve the AP for large objects by 1.4% (from 52.2% to 53.6%), which is much higher than the improvement using conventional convolutions (*i.e.*, 1.4% vs. 0.1%). It demonstrates that our center pooling is effective in detecting center keypoints of objects, especially for large objects. Our explanation is that center pooling can extract richer internal visual patterns, and larger objects contain more accessible internal visual patterns. Figure 6(e) shows the results of detecting center keypoints without/with center pooling. We can see the conventional convolution fails to locate the center keypoint for the cow, but with center pooling, it successfully locates the center keypoint.

Cascade corner pooling. We replace corner pooling [20] with cascade corner pooling to detect corners (see CCP in the table). The second row in Table 4 shows the results that we test on the basis of CornerNet511-52. We find that cascade corner pooling improves the AP by 0.7% (from 37.6% to 38.3%). The last row shows the results that we test on the basis of CenterNet511-52, which improves the AP by 0.5% (from 40.8% to 41.3%). The results of the second row show there is almost no change in the AP for large objects (*i.e.*, 52.2% vs. 52.2%), but the AR is improved by 1.8% (from 74.0% to 75.8%). This suggests that cascade corner pooling can “see” more objects due to the rich internal visual patterns in large objects, but too rich visual patterns may interfere with its perception for the boundary information, leading to many inaccurate bounding boxes. After equipping with our CenterNet, the inaccurate bound-

ing boxes are effectively suppressed, which improves the AP for large objects by 2.2% (from 53.6% to 55.8%). Figure 6(f) shows the result of detecting corners with corner pooling or cascade corner pooling. We can see that cascade corner pooling can successfully locate a pair of corners for the cat on the left while corner pooling cannot.

4.6. Error Analysis

The exploration of visual patterns within each bounding box depends on the center keypoints. In other words, once a center keypoint is missed, the proposed CenterNet would miss the visual patterns within the bounding box. To understand the importance of center keypoints, we replace the predicted center keypoints with the ground-truth values and evaluate performance on the MS-COCO validation dataset. Table 5 shows that using the ground-truth center keypoints improves the AP from 41.3% to 56.5% for CenterNet511-52 and from 44.8% to 58.1% for CenterNet511-104, respectively. APs for small, medium and large objects are improved by 15.5%, 16.5%, and 14.5% for CenterNet511-52 and 14.5%, 14.1%, and 13.3% for CenterNet511-104, respectively. This demonstrates that the detection of center keypoints is far from the bottleneck.

5. Conclusions

In this paper, we propose CenterNet, which detects objects using a triplet, including one center keypoint and two corners. Our approach addresses the problem that CornerNet lacks an additional look into the cropped regions by exploring the visual patterns within each proposed region with minimal costs. In fact, this is a common defect for all one-stage approaches. As one-stage approaches remove the RoI extraction process, they cannot pay attention to internal information within cropped regions.

An intuitive explanation of our contribution lies in that we equip a one-stage detector with the ability of two-stage approaches, with an efficient discriminator being added. We believe that our idea of adding an extra branch for the center keypoint can be potentially generalized to other existing one-stage approaches (*e.g.*, SSD [27]). Meanwhile, some advanced training strategies [46] can be used for better performance. We leave as our future work.

CRE	CTP	CCP	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L	AR ₁	AR ₁₀	AR ₁₀₀	AR _S	AR _M	AR _L
			37.6	53.3	40.0	18.5	39.6	52.2	33.7	52.2	56.7	37.2	60.0	74.0
		✓	38.3	54.2	40.5	18.6	40.5	52.2	34.0	53.0	57.9	36.6	60.8	75.8
✓			39.9	57.7	42.3	23.1	42.3	52.3	33.8	54.2	58.5	38.7	62.4	74.4
✓	✓		40.8	58.6	43.6	23.6	43.6	53.6	33.9	54.5	59.0	39.0	63.2	74.7
✓	✓	✓	41.3	59.2	43.9	23.6	43.8	55.8	34.5	55.0	59.2	39.1	63.5	75.1

表4: 在MS-COCO验证数据集上对CenterNet511-52主要组件进行的消融研究。CRE表示中心区域探索, CTP表示中心池化, CCP表示级联角点池化。

Method	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
CenterNet511-52 w/o GT	41.3	59.2	43.9	23.6	43.8	55.8
CenterNet511-52 w/ GT	56.5	78.3	61.4	39.1	60.3	70.3
CenterNet511-104 w/o GT	44.8	62.4	48.2	25.9	48.9	58.8
CenterNet511-104 w/ GT	58.1	78.4	63.9	40.4	63.0	72.1

表5: 通过使用真实值对中心关键点进行误差分析。我们将预测的中心关键点替换为真实值, 结果表明在检测中心关键点方面仍有改进空间。

比大型物体的位置更精确。

中心池化。为了证明所提出的中心池化的有效性, 我们在网络中加入了中心池化模块 (见表格中的CTP)。表4的第四行显示, 中心池化将AP提高了0.9% (从39.9%提升至40.8%)。值得注意的是, 借助中心池化, 我们将大型物体的AP提升了1.4% (从52.2%提升至53.6%), 这一提升幅度远高于使用传统卷积的改进 (*i.e* 1.4%对0.1%)。这表明我们的中心池化能有效检测物体的中心关键点, 尤其对大型物体效果显著。我们的解释是, 中心池化能提取更丰富的内部视觉模式, 而大型物体包含更多可获取的内部视觉模式。图6(e)展示了未使用/使用中心池化检测中心关键点的结果。可以看到, 传统卷积未能定位奶牛的中心关键点, 但加入中心池化后, 成功定位了其中心关键点。

级联角点池化。我们将角点池化[20]替换为级联角点池化以检测角点 (见表格中的CCP)。表4第二行展示了我们在CornerNet511-52基础上测试的结果。我们发现级联角点池化将AP提升了0.7% (从37.6%提升至38.3%)。最后一行展示了我们在CenterNet511-52基础上测试的结果, 其将AP提升了0.5% (从40.8%提升至41.3%)。第二行的结果显示, 大目标的AP几乎无变化 (*i.e* 52.2% vs. 52.2%), 但AR提升了1.8% (从74.0%提升至75.8%)。这表明级联角点池化能凭借大目标丰富的内部视觉模式“看到”更多目标, 但过于复杂的视觉模式可能干扰其对边界信息的感知, 导致产生大量不准确的边界框。在配备我们的CenterNet后, 这些不准确的边界——

检测框被有效抑制, 这使大型物体的AP提高了2.2% (从53.6%提升至55.8%)。图6(f)展示了使用角点池化或级联角点池化检测角点的结果。可以看出, 级联角点池化能成功定位左侧猫咪的一对角点, 而角点池化则无法实现。

4.6. 误差分析

每个边界框内视觉模式的探索依赖于中心关键点。换言之, 一旦中心关键点未被检测到, 所提出的CenterNet就会遗漏边界框内的视觉模式。为理解中心关键点的重要性, 我们将预测的中心关键点替换为真实值, 并在MS-COCO验证数据集上评估性能。表5显示, 使用真实中心关键点将CenterNet511-52的AP从41.3%提升至56.5%, 将CenterNet511-104的AP从44.8%提升至58.1%。对于小、中、大目标, CenterNet511-52的AP分别提升了15.5%、16.5%和14.5%, CenterNet511-104则分别提升了14.5%、14.1%和13.3%。这表明中心关键点的检测远未成为性能瓶颈。

5. 结论

本文提出CenterNet, 它使用一个三元组检测物体, 包括一个中心关键点和两个角点。我们的方法解决了CornerNet缺乏对裁剪区域额外观察的问题, 以最小成本探索每个提议区域内的视觉模式。事实上, 这是所有单阶段方法的共同缺陷。由于单阶段方法移除了RoI提取过程, 它们无法关注裁剪区域内的内部信息。

我们贡献的一个直观解释在于, 我们为单阶段检测器赋予了双阶段方法的能力, 同时添加了一个高效的判别器。我们相信, 这种为中心关键点添加额外分支的思路, 有可能推广到其他现有的单阶段方法 (*e.g*, 如SSD [27])。同时, 一些先进的训练策略[46]可用于提升性能。我们将此作为未来的工作方向。

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